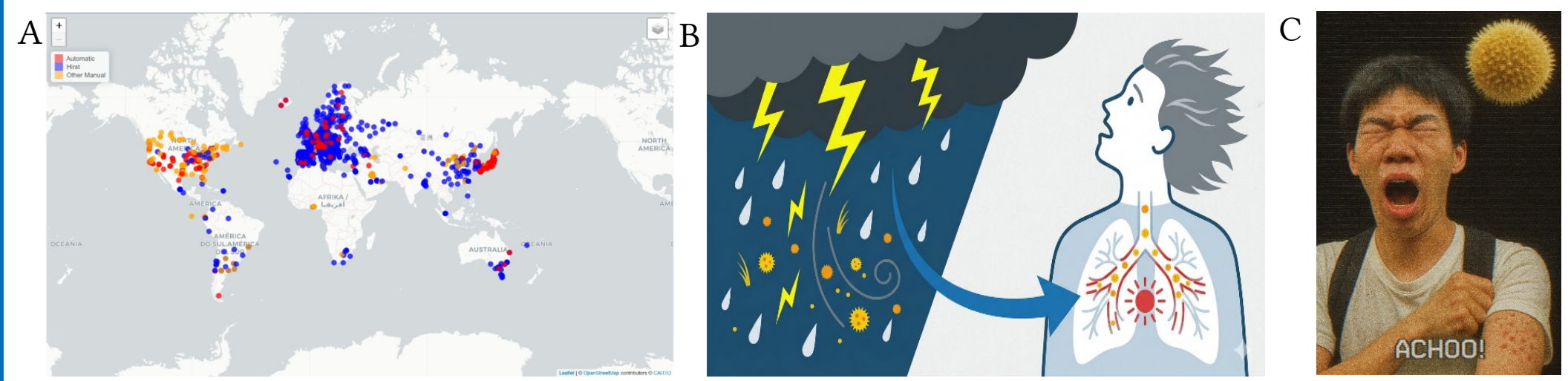


Introduction

Pollen Allergy: A Global Public Health Issue



Pollen allergies have become an increasingly serious **global public health issue**. In every **four** people, **one** is allergic to pollen¹. They experience symptoms from **allergic rhinitis** to **life-threatening anaphylactic shocks**, and heavily depend on up-to-date pollen counts in their area.

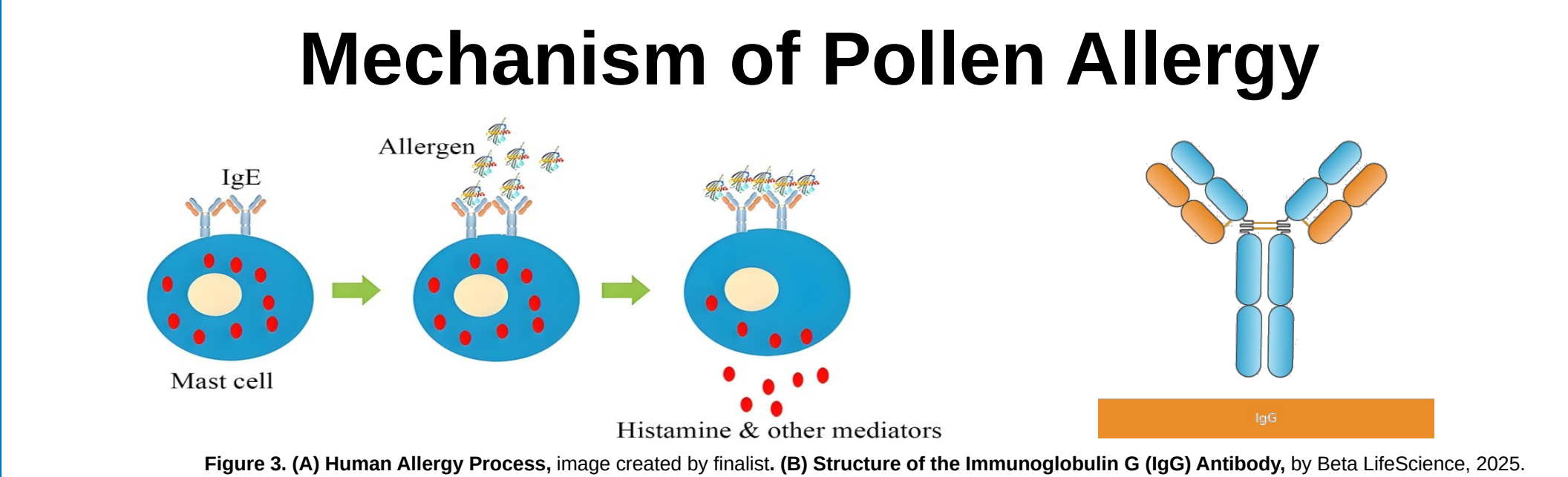
Current Pollen Allergen Detecting Methods

- However, traditional pollen detection methods are:
 - Too Slow** (count up to ~4 hrs) → **daily pollen count rely on data from the previous day**.
 - Not Precise** (detects pollen grains instead of proteins) → **vulnerable to Thunderstorm Asthma²**, where pollen proteins surge while the grain vanished.
- Too Costly** (**AI-based** pollen image recognition microscope **cost ~\$23k per device**)

Research Goal

Enable accurate, real-time airborne pollen detection

Detection Principles



- Human Allergy & IgG Recognition**
 - Allergic response:**
 - Triggered when allergens (e.g., pollen proteins) are recognized by the immune system.
 - Primarily mediated by IgE antibodies, which leads to inflammatory reactions.
 - IgG antibody:**
 - Fab region binds specifically to antigen epitopes → **High specificity of antigen-antibody binding**.
 - This enables the sensor to selectively **recognize** and **differentiate** pollen proteins from impurities, which is the fundamental principle of this study.

Graphene Properties and Doping

Two-dimensional structure & Large surface area & High carrier mobility

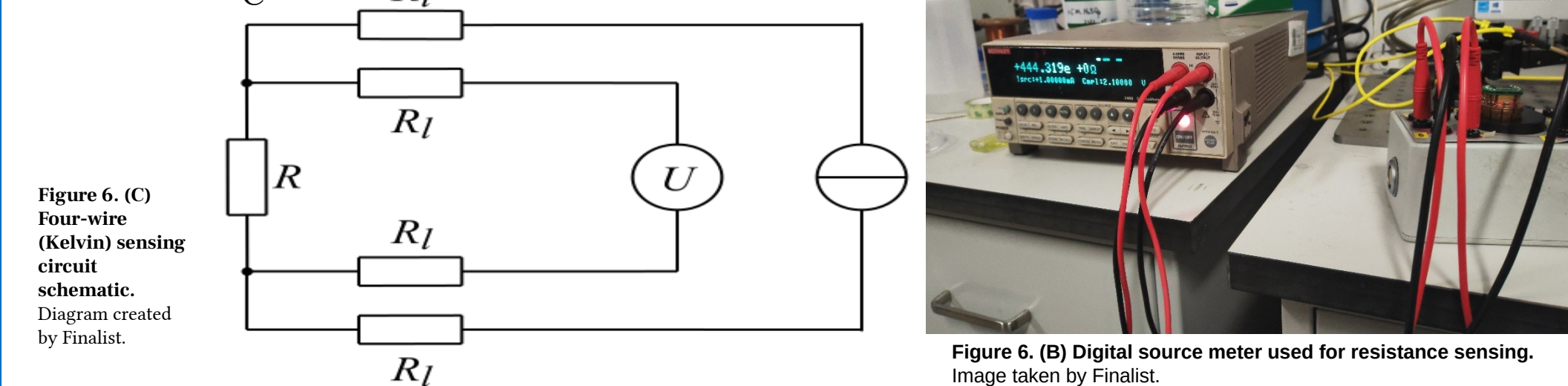
- High electrical conductivity**, enabling ultra-sensitive detection of biomolecular binding and gas adsorption
- Detects **subtle surface interactions** (e.g., antigen binding, doping)

Doping Mechanism

- Impurities land on graphene surface → Electron withdrawal → **increases hole carrier density (p-type)**
- Leads to a **measurable** increase in electrical resistance and a **shift in Dirac Point**.
- IgG functionalization & antigen binding induce **p-type*** doping in graphene, shifting the **Fermi level downward and the Dirac point rightward**.
- Pollen protein binding increases graphene resistance through **physical blocking** (carrier scattering) and **charge transfer**. Since pollen proteins are typically positively charged near neutral pH (~7.0), the observed resistivity change is mainly attributed to **specific antigen-antibody interactions** rather than non-specific adsorption.

Graphene Four-Wire Sensing

- Due to graphene's low intrinsic resistance, contact resistance between the electrodes and graphene can exceed the graphene resistance itself.
- Therefore, **Four-wire electrodes** were used to **minimize contact resistance** and achieve **Lower Limit of Detection**.



Methodology

Graphene Chemical Modification⁵: PBASE Linker, IgG Antibody and Pollen Allergen

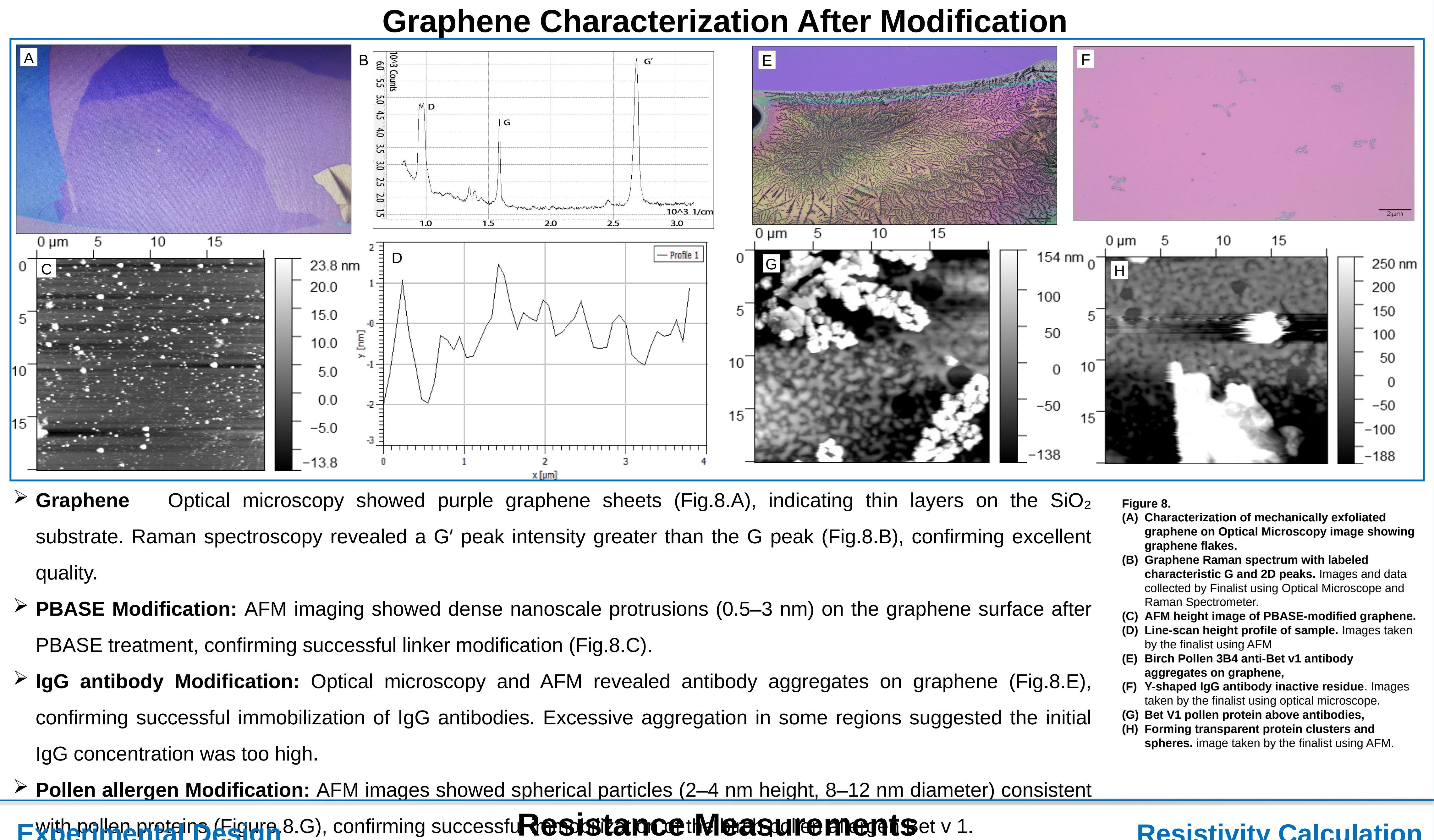
- Sensing interface construction**
 - Graphene-PBASE-IgG functional structure for antigen capture
- PBASE linker** ($C_{24}H_{19}NO_4$):
 - Pyrene group → $\pi-\pi$ stacking with graphene
 - NHS ester group → covalent binding to IgG antibodies³
- Antibody functionalization:**
 - IgG antibodies introduced in PBS solution
 - Ragweed, Birch IgG (1:5, 1:10, 1:15 dilution) immobilized on graphene

Forms a selective sensing interface for pollen antigen binding

Step	Interaction
Pyrene ↔ Graphene	$\pi-\pi$ stacking
NHS ester ↔ IgG	Covalent amide bond

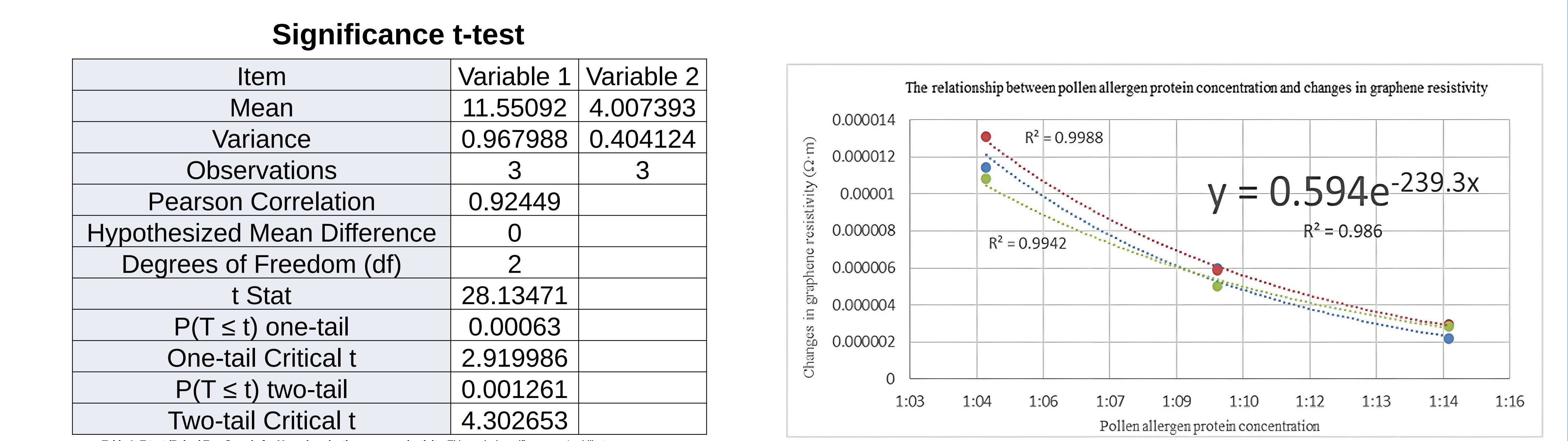
Figure 7: (A) Chemical structure of 1-Pyrenebutyric acid NHS ester (PBASE). Graphic created on April 7, 2025, using ChatGPT-5 by OpenAI. (B) Graphene-PBASE-IgG antigen sensing interface. Graphic created by Finalist using Microsoft PowerPoint.

Results

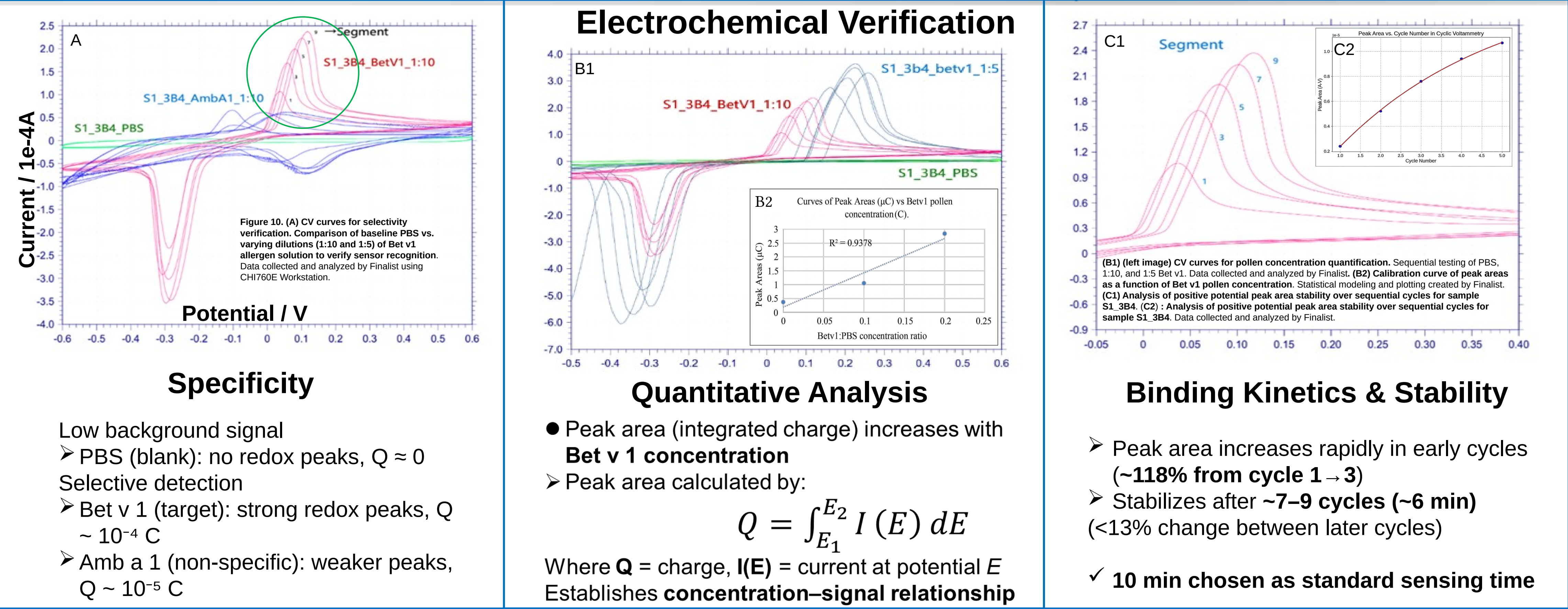


Experimental Design

- Group 1 (Selectivity Test)**
 - Antibody-functionalized graphene exposed to: **Specific** vs **non-specific** pollen proteins.
 - Measure **resistance change (ΔR)** before/after binding.
- Group 2 (Quantification Test)**
 - Sensor exposed to **varying concentrations** (1:5, 1:10, 1:15) of target pollen proteins.
 - Measure **ΔR vs concentration** relationship.



At $\alpha = 0.05$, a one-tailed t-test shows post-doping resistivity change is significantly greater than pre-doping, confirming the doping effect.



Discussion & Conclusion

- Sensitivity**
 - limit of detection (LOD): (graphene sensor ~38.6 $\mu g/mL$) Detects $\Delta\rho$ down to **$2.16 \times 10^{-6} \Omega \cdot m$ (1:15 dilution)** Below typical immunization dose (150 μg protein)
- Specificity**
 - Specific vs non-specific response: **2.6× higher signal** – confirms target-specific detection (signal-to-noise ratio)
- Cost**
 - Graphene sensor (<\$70 per device) VS ML-based pollen image recognition microscope (>\$160k) VS electrochemical workstations (>\$4k)
- Timing**
 - Reaction stabilizes at **~200 s**, which shows binding kinetics and optimal time window
 - **10 min chosen as standard sensing time**

Implications of Exponential Relationship

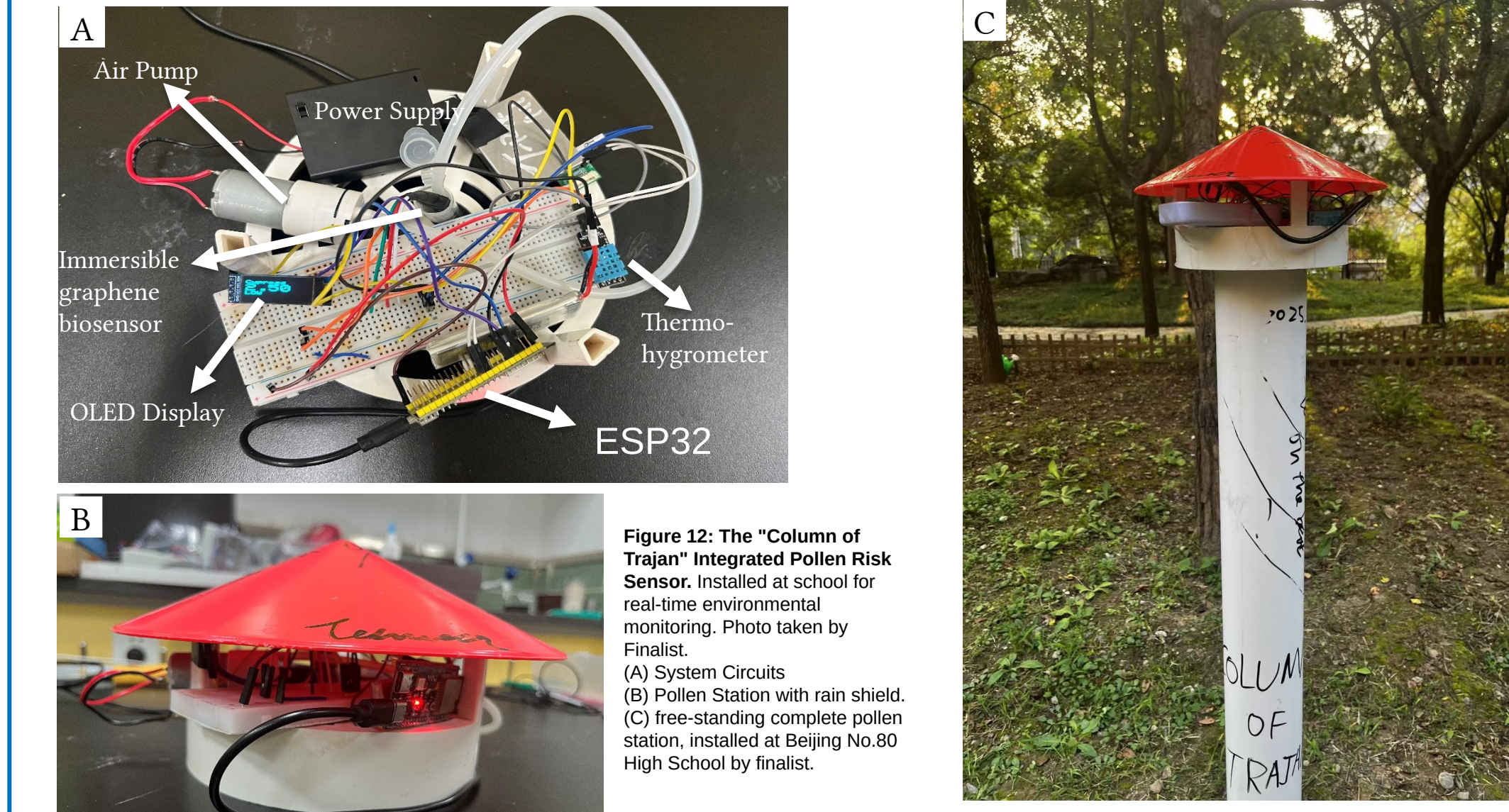
- Exponential relationship ($R^2>0.98$)** between pollen concentration and resistivity:
- Indicates signal is governed by **specific antigen-antibody binding**
 - Consistent with **binding-site saturation behavior**.
 - Helps achieve **quantitative detection over extremely wide concentration range**.

This study's chemically modified graphene biosensors achieve selective, quantitative detection of airborne allergen proteins.

Limitations and Future Outlooks

- Mechanically exfoliated and CVD-grown graphene showed variability in electrical conductivity, limiting sensor consistency.
- The graphene sensors were only tested with **three common allergens in Beijing**(Ragweed Amb A1, Birch Bet v1, Cypress Cup a 1). Future work should expand detection to additional allergen species.
- Broad diagnostic potential:** The same sensor could perform human allergy detection by detecting human IgG using antigen-functionalized graphene, providing a safer alternative for the risky Skin Prick Test.

Engineering Adaptation



- Portable system integration:** The graphene biosensor was combined with environmental sensors & ESP32 & AD5940 chip, achieving rapid ~10-minute testing workflow. It was also integrated with other environmental sensors(e.g.**humidity sensor**) to calculate the overall allergy risk.
- The station now provides daily health advisories to 1,800 students teaching staff at Beijing No.80 High School and has increased students' attendance rate during allergy season by 60%.**

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